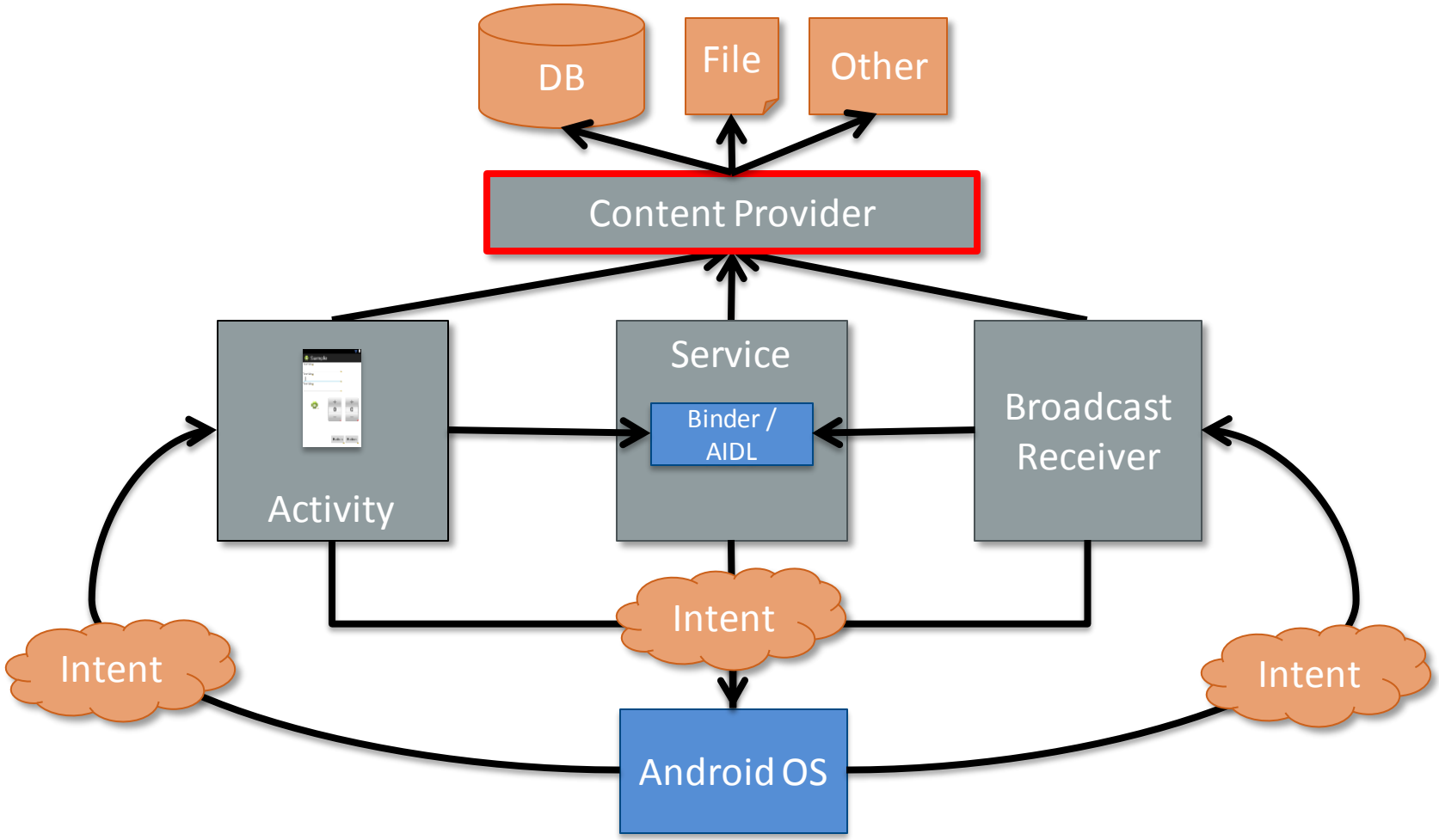
605.486 – Android Development

Module 7a
Content Providers

Application Components

and Typical Communication



Content Providers

- Expose data to application components
  - Can be private to your application (export=false)
  - Can be available to other applications (export=true)
  - REST-like interface
- Often uses SQLite to store/access data
  - BTW: SQL is properly pronounced “Squeal”

Content Provider Basics (1)

- Constants useful for table / column names
- ID column must be named "_id" for use with SimpleCursorAdapter
- Database Helper
  - If db not found -> calls onCreate()
  - If db found, version unchanged, no action
  - If db found, version changed -> onUpgrade()

Content Provider Basics

- onCreate()
  - Create helper
  - call getWriteableDatabase()
  - store in field
- NOTE: There is no way to close the DB!
  - This is ok – you may see warnings when exiting app, though
  - Database is safe on each write

Content URIs

- content://com.javadude.todo/todo
- content://com.javadude.todo/todo/42
- We need to be able to determine what user wants
  - UriMatcher matches the URI and returns a code

Content Provider - query

- query() method fetches data by URI & parameters
  - uri – identifies “directory” or specific item
  - projection – specifies columns to return
  - selection – specifies columns for search
  - selectionArgs – values for specified columns
  - sortOrder – requested result order

- Returns a cursor
  - *Caller must close cursor!*
  - Preferably use a CursorLoader and it will take care of closing

Content Provider - query

- URI Matcher to determine type of request
- Call query() on the database to get cursor
- Set notification Uri on the cursor
  - *This is extremely important!!!*
  - Tells cursor which URI represents its data so it can listen for changes and refresh if needed
- Return cursor

Content Provider - getType

- Return MIME type of data for a URI
- Use URI Matcher to determine type of URI
- Typically you have
  - Content Directory
  - Content Item
- May be more than one directory / item type
  - Often there is only one of each

Content Provider - insert

- Call insert() on the database
- DB insert returns ID of inserted item
- Create Uri with new ID
- Notify data source has changed
  - using inserted Uri
- Return the Uri

Content Provider - delete

- Call delete() on the database
  - Returns the number of deleted rows
- Notify data source has changed
  - using specified Uri
- Return number of deleted rows

Content Provider - update

- Call update() on the database
  - Returns number of updated rows
- Notify data source has changed
  - using specified Uri
- Return number of updated rows

Content Provider - Registration

- Register provider in the manifest

```
<provider android:name=".TodoContentProvider"
        android:authorities="com.javadude.todo"
        android:exported="false"/>
```

Content Provider - Exporting

- Export if desired
  - If exported, should specify permission!
  - Use existing permissions, or create new ones
  - New permissions:
    - "normal" – other apps must specify permission; user not notified
    - "dangerous" – user will be notified
    - "signature" – only other apps with same signature can use

Using the Content Provider

- Create Uri
  - Uri.withAppendedPath(CP.CONTENT_URI, stuffToAppend)
- Talk to content resolver
  - Context.getContentResolver()
  - Locates registered provider for authority
  - Call query, update, delete, insert

CursorLoader Usage

- Runs query in background during startup

- Create SimpleCursorAdapter for ListView
- Specify query parameters
- Swap new cursor into SimpleCursorAdapter when finished
- Restart loader to refresh contents
  - Usually not needed if notification used!